Task 11: Data Access Layer using Spring Data JPA

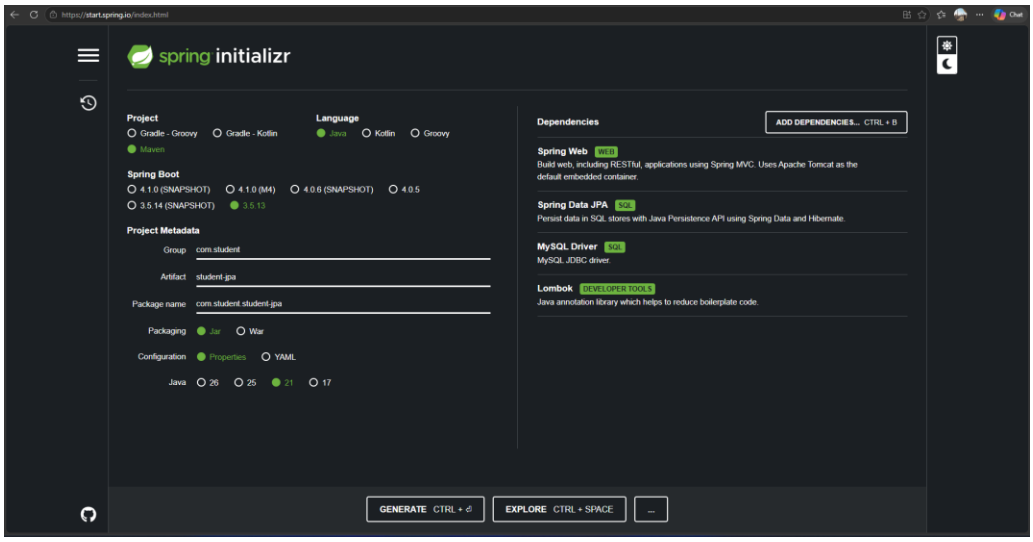
Aim

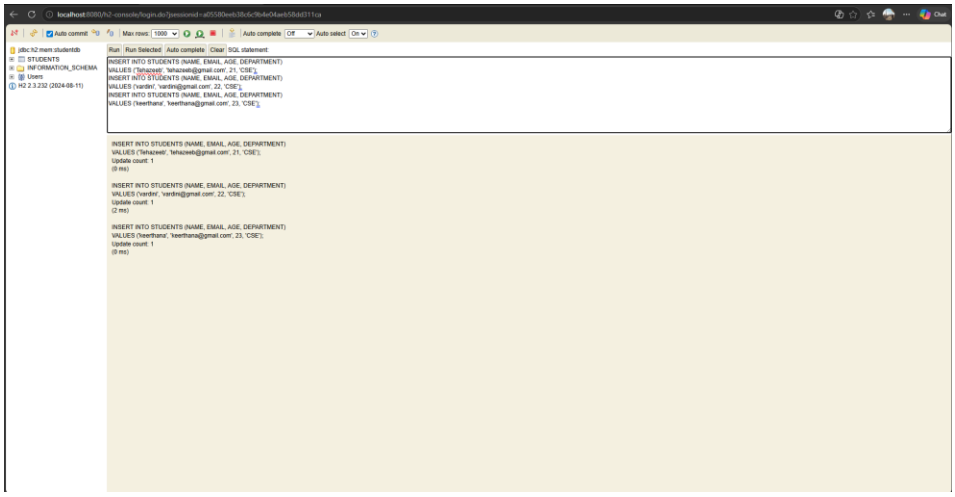
To implement a **Data Access Layer** using **Spring Data JPA** by using JpaRepository and custom query methods to retrieve student records based on conditions such as **department** or **age**, and also support **sorting** and **pagination** for efficient data access.

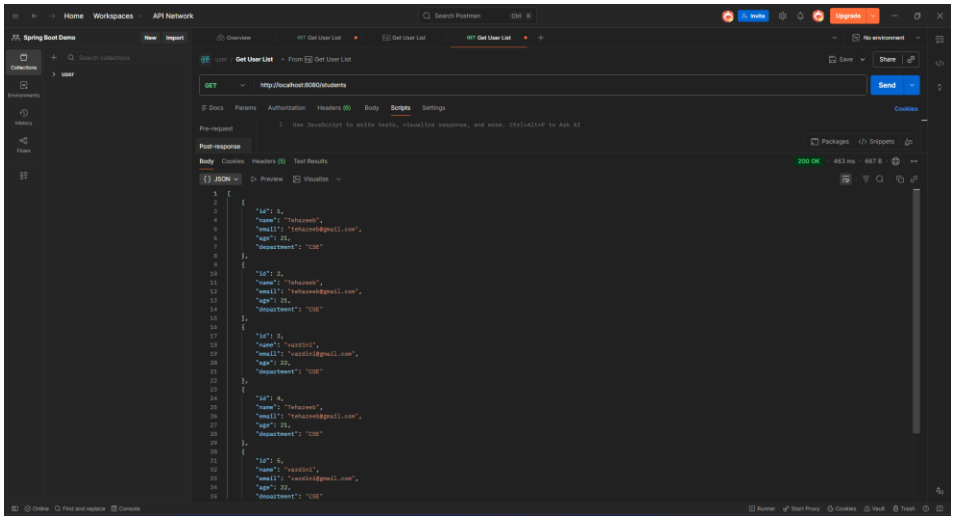
Algorithm

1. Start the application.
2. Create a Spring Boot project with required dependencies.
3. Configure database connection in application.properties.
4. Create a Student entity class and map it to the database table.
5. Create a repository interface that extends JpaRepository.
6. Add custom query methods to retrieve student records by department and age.
7. Add methods for sorting records by name, age, or department.
8. Add pagination support using Pageable.
9. Create a service class to handle business logic.
10. Create a controller class to expose API endpoints.
11. Insert sample records into the database.
12. Fetch records based on conditions, sorting, and pagination.
13. Display the output.
14. Stop the application.

Outputs:







Result:

To implement a **Data Access Layer** using **Spring Data JPA** by using JpaRepository and custom query methods to retrieve student records based on conditions such as **department** or **age**, and also support **sorting** and **pagination** for efficient data access is executed successfully.